


$$\text{Slope} = \frac{7-4}{2-0}$$

$$\text{Slope} = 3/2$$

① Problem understanding

↓

② Data understanding

↓

③ Extract X, Y

↓

④ Data cleaning (Cleaning is not part of EDA mostly, But also some time consider as EDA).

↓

⑤ EDA

↓

⑥ Feature scaling

↓

⑦ Encoding

↓

⑧ Train test split

↓

⑨ Train data using algorithm

↓

⑩ Check the accuracy

17-09-2026 (Algorithms) → Similar to function, whose having a predefined logic, helps to train data with

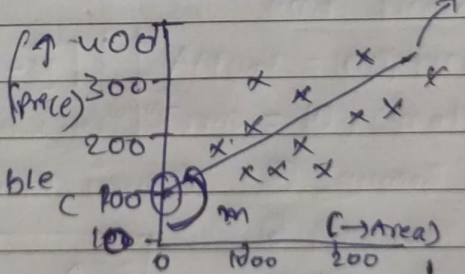
① Linear Regression:- $y = mx + c$

y = Target Variable

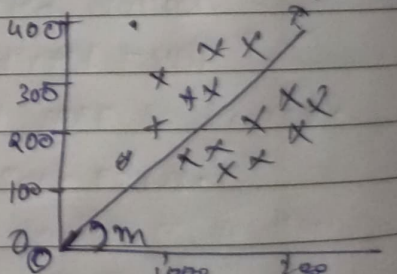
m = Slope

x = Input Variable

c = y-Intersect




Best fit line




Best fit line

m are random values not angles.

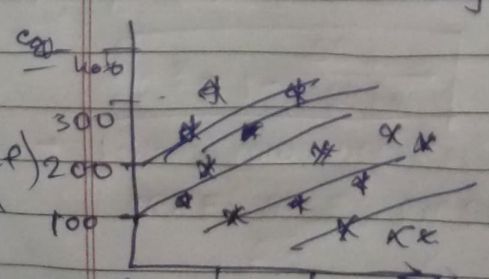
$c = 100$; $m = 135$

If $x = 25$; $y = \underset{m}{135}\,\underset{x}{(25)} + \underset{c}{100}$

$c = 0$; $m = 45$

$y = \underset{m}{45}\,\underset{x}{(25)} + \underset{c}{0}$

Best Fit line:- In linear regression model, we can have many lines, How we decide best among them.




For each fit line different m(s) c values. So, model calculates the Total Error for each fit line.

Residual Error = Actual - Predicted

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$ slope formulae.

already known Data.

Eg: If we take line with: $m = 0.5$, $c = 100$ & $x = 500$

$y = 0.5(500) + 100 \Rightarrow \boxed{350}$ //

350 → Calculated

120 → actual

difference ⇒ $\boxed{\text{Error}_{(E)} = 350 - 120 = 230}$

If calculate Error for all known data points (E) Sum them all.

$\boxed{\text{Total Error} = E1 + E2 + \cdots En}$ ⇒ for Fit line with $m = 0.5$, $c = 100$.

Continuously, calc: Total Error for all fit lines (E)

Finally,

Fit line with minimum total Error value is considered as Best fit line.

Sometimes, total Error value looks same for 2(3) models but, if we consider exact values, we will eliminate line with slightly higher Error value

Note: $\boxed{\text{Finally, Linear Regression model saves } m, c \text{ values of Best fit line.}}$

Definition: Linear regression assumes linear relationship b/w input variables (E) output variable s. It finds Best fit line, to predict continues numerical value.

→ Linear relationship, If input variable increases (↑) output " " (↑)

(X α Y) If, X (↓), Y (↓).

2 types of regression:

① Simple Linear Regression: one independent variable predicts one dependent variable. (Eg: Area (X) → price (Y))

② Multiple linear Regression: multiple independent variable predicts one dependent variable.

Eg: Rooms (multiple X), Bathroom, Ratings → Rent of an house (Y)

→ Optimization: finding Best fit line which has minimum Error value, better accuracy.

① Simple linear regression: $\boxed{y = b_0 + b_1 x}$ ($y = c + mx$)

② multiple linear regression: $\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3 + \cdots$

Cost Function: $\boxed{MSE = \frac{1}{n}\sum (y - \hat{y})^2}$

classmate
Date
Page

★★★★ 5 Key assumptions of Linear Regression:-

→ ① Linearity:- The relationship b/w Independent Variable (X) & dependent Variable (Y) should be approximately linear.

→ ② Independence:- Each observation should be independent of the others

Eg:- One House prediction /error should not depend on another house's observation.

→ ③ Homoscedasticity:- The Spread (Variance) of the residual (errors) should be roughly the same across all levels (small, medium, large inputs) of your predictor variable

→ ④ Normality:- Residuals should be approximately normally distributed

→ ⑤ NO multicollinearity:- Independent Varibles should not be highly correlated.

(L-I-H-N-N) Linearity → Independence → Homoscedasticity → Normality → No multicollinearity

★★★ Evaluation Metrics for Regression:- After training, we evaluate how well model performs.

① MAE (Mean Absolute Error):- Average absolute difference actual and predicted values.

$$MAE = \frac{1}{n} \Sigma |y - \hat{y}|$$

(Actual value, predicted value)

Actual = 100, Predicted = 90, Error = 10, small MAE is better.

② MSE (Mean Squared Error):- ⊙ Squared Each error.
⊙ penalizes larger mistakes heavily.
⊙ smaller MSE is better.

$$MSE = \frac{1}{n} \Sigma (y - \hat{y})^2$$

③ RMSE (Root Mean Squared Error):- $RMSE = \sqrt{MSE}$

→ converts MSE back in to the original unit.

④ $R^2$ score (Coefficient of Determination):-

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Range:- 0 → Poor model
1 → Perfect model

0.75 - Good
0.50 - Average
0.25 - Below avg

→ $R^2$ tells how much variation in the target variable is explained by model

MAE, MSE, RMSE → Lower (Best Value)
$R^2$ → Higher (Best Value)

(2.) Classification:- Predicts categories (or) class labels.

Types:-

→ (1.) Binary Classification:- Two possible classes as outcome.
output:- 0 (or) 1 . Eg:- pass / Fail, spam / Not spam, Disease / No Disease.

→ (2.) multi-class classification:- Three (or) more classification
Eg:- Movie rating, Animal classification etc...
output:- Cat, Dog, Horse.

→ (3.) multi-label classification:- one instance can belong to multiple class simultaneously.
An article may have tags: AI, Python, machine learning.

(*)(*) Logistic Regression:- predicts, probability that input belongs to a class.

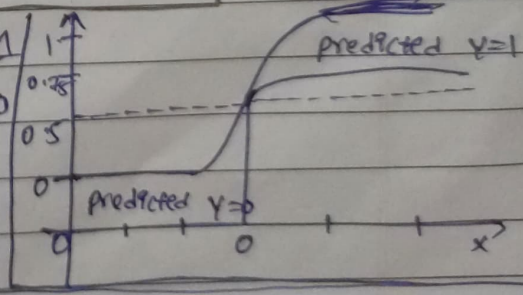
→ Although it contains "Regression", it is classification algorithm.

Eg:- Email → spam probability = 0.92 → spam.

Sigmoid Function:- $p = \frac{1}{1 + e^{-(b_0 + b_1 x)}}$ output ranges from 0 and 1.

Eg, Probability ≥ 0.5 → class 1
Probability < 0.5 → class 0

→ S-shaped curve

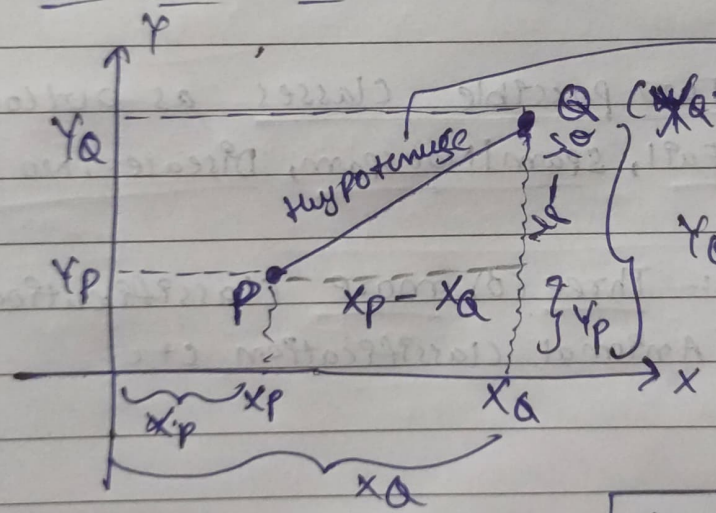



| Linear Regression | Logistic Regression |
|---|---|
| (1.) used for regression | (1.) used for classification |
| (2.) predicts continuous numerical value | (2.) predicts a class / category |
| (3.) uses a straight line | (3.) uses S-shaped (sigmoid) curve |
| (4.) o/p can be any numeric value | (4.) o/p is probability b/w 0 → 1 |
| (5.) Common metrics: MAE, MSE, RMSE, $R^2$ | (5.) Accuracy, precision, Recall, F1 |
| (6.) No threshold value used | (6.) 0.5 threshold used. |
| Equation: $y = b_0 + b_1 x$ | used sigmoid function. |

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Date
Page

# Error Finding with

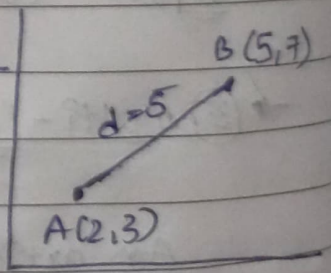
(***) Euclidean distance:- Shortest distance between 2 datapoints




$$d(P,Q) = \sqrt{(X_P - X_Q)^2 + (Y_P - Y_Q)^2}$$

Eg: A (2,3)
B (5,7)

$$d = \sqrt{(5-2)^2 + (7-3)^2}$$
$$d = \sqrt{3^2 + 4^2}$$
$$d = \sqrt{25} = 5$$




→ Euclidean distance mostly used in ~~K-neighbours~~ K-Nearest Neighbours (KNN), clustering etc...